MILESTONE 2 • 3 MARKS

Project Requirements Presentation

JAVA DESKTOP APPLICATION

Swing UI • JDBC • Database

Define the problem. Agree on what the system must do.

PROJECT TITLE

[Enter project title]

Team No.: [ ] | Class/Semester: [ ] | Review Date: [ ]

MANDATORY EVIDENCE

Team and HackerRank Java progress

Update this table in every milestone submission and PPT

#	Student name	Roll no.	Role	Profile / username	Previous Java stars	Current Java stars	Progress
1	[Name]	[Roll no.]	**Team Lead**	[Profile link]	[0–5]	**[0–5]**	**[+ / —]**
2	[Name]	[Roll no.]	Member	[Profile link]	[0–5]	**[0–5]**	**[+ / —]**
3	[Name]	[Roll no.]	Member	[Profile link]	[0–5]	**[0–5]**	**[+ / —]**
4	[Name]	[Roll no.]	Member	[Profile link]	[0–5]	**[0–5]**	**[+ / —]**
5	[Name]	[Roll no.]	Member	[Profile link]	[0–5]	**[0–5]**	**[+ / —]**

REQUIRED

Show the profile URL or username and record Java stars as on the review date. Every member must demonstrate continuing Java practice.

Recommended evidence: a current HackerRank profile screenshot in the appendix or profile link in this table.

The problem and its context

Make the need for the application clear before describing features

EXISTING PROCESS

[How is the work currently done?]

[Manual tools, files or steps used]

AFFECTED USERS

[Who experiences the problem?]

[Primary and secondary users]

PAIN POINTS

[Delay / errors / duplication]

[Why the current method is inadequate]

PROBLEM STATEMENT

[In 2–3 sentences, state the specific problem, the users affected, and the consequence. Avoid describing the solution here.]

Objectives and project scope

Separate expected outcomes from features and implementation choices

PROJECT OBJECTIVES

01 [Objective 1 — user or process outcome]

02 [Objective 2 — data or efficiency outcome]

03 [Objective 3 — quality or reporting outcome]

04 [Objective 4 — learning / technical outcome]

SCOPE BOUNDARY

IN SCOPE

- [Feature / operation]
- [User or module]
- [Report / data task]
- [Required validation]

OUT OF SCOPE

- [Excluded feature]
- [Future enhancement]
- [External integration]
- [Known limitation]

Keep the scope achievable within the mini-project schedule.

Users, responsibilities and system flow

Show who performs each major operation and how data moves through the application

USERS / ACTORS

01 **[Primary user]**
[Main activities]

02 **[Secondary user]**
[Main activities]

03 **[Administrator / other]**
[Main activities]

IMPROVEMENT OVER CURRENT PROCESS

[State the measurable improvement: faster, fewer errors, better traceability, etc.]

SYSTEM FLOW



PROPOSED SOLUTION — 3 TO 4 SENTENCES

[Explain the application purpose, intended users, major operations, and how it addresses the stated problem.]

NO DESIGN DETAILS YET

What the system shall do

Use clear, testable statements; one important function per row

Req. ID	Requirement statement	Actor	Priority	Acceptance check
FR-01	The system shall [perform one observable function]	[User]	H / M / L	[How will you verify it?]
FR-02	The system shall [perform one observable function]	[User]	H / M / L	[How will you verify it?]
FR-03	The system shall [perform one observable function]	[User]	H / M / L	[How will you verify it?]
FR-04	The system shall [perform one observable function]	[User]	H / M / L	[How will you verify it?]
FR-05	The system shall [perform one observable function]	[User]	H / M / L	[How will you verify it?]
FR-06	The system shall [perform one observable function]	[User]	H / M / L	[How will you verify it?]
FR-07	The system shall [perform one observable function]	[User]	H / M / L	[How will you verify it?]
FR-08	The system shall [perform one observable function]	[User]	H / M / L	[How will you verify it?]

QUALITY CHECK **Specific • Necessary • Feasible • Testable • Traceable to a user or module**

Major modules and proposed Swing screens

Group related requirements into manageable parts of the application

M1 [Module name]

Core functions
[Function 1]
[Function 2]

Swing screen(s): [ ]

Related FRs: [FR-__ , FR-__]

M2 [Module name]

Core functions
[Function 1]
[Function 2]

Swing screen(s): [ ]

Related FRs: [FR-__ , FR-__]

M3 [Module name]

Core functions
[Function 1]
[Function 2]

Swing screen(s): [ ]

Related FRs: [FR-__ , FR-__]

M4 [Module name]

Core functions
[Function 1]
[Function 2]

Swing screen(s): [ ]

Related FRs: [FR-__ , FR-__]

Show clear ownership: every functional requirement should belong to a module.

Database and input/output requirements

Identify what must be stored and validated—detailed schema comes in the design milestone

Entity / data	Key information to store	Relationship / owner	CRUD needed
[Entity 1]	[Important fields]	[Related entity / user]	C / R / U / D
[Entity 2]	[Important fields]	[Related entity / user]	C / R / U / D
[Entity 3]	[Important fields]	[Related entity / user]	C / R / U / D
[Entity 4]	[Important fields]	[Related entity / user]	C / R / U / D

INPUT VALIDATION

- Mandatory fields
- Data type and range
- Date / format checks
- Duplicate prevention
- Meaningful error messages

IMPORTANT INPUTS

[Forms, search terms, transaction data, filters]

EXPECTED OUTPUTS

[Tables, confirmations, reports, summaries, errors]

Non-functional requirements and technology constraints

Turn quality expectations into measurable acceptance conditions where possible

USABILITY

[Simple navigation / readable forms]

RELIABILITY

[Correct save / retrieve behaviour]

PERFORMANCE

[Expected response time / data size]

VALIDATION

[Rejected inputs and checks]

ERROR HANDLING

[User message + safe recovery]

MAINTAINABILITY

[Naming, modular code, JavaDocs]

MANDATORY TECHNOLOGY BASELINE

LANGUAGE	DESKTOP UI	CONNECTIVITY	DATABASE	IDE
Java	Swing	JDBC	[Name / version]	[Name / version]

REVIEW SUMMARY

Requirements baseline

Confirm the scope before moving to project design

MILESTONE EVIDENCE CHECKLIST

- ✓ Problem, users and objectives are clearly stated
- ✓ In-scope and out-of-scope items are agreed
- ✓ Functional requirements are testable and prioritised
- ✓ Modules, inputs, outputs and stored data are identified
- ✓ Non-functional requirements and technology choices are stated
- ✓ Requirements document and PPT use the same scope

DECISION REQUIRED

[Approved / Revise]

Key changes requested:
[_____]

NEXT MILESTONE

Project Design Document & PPT

Carry forward the updated HackerRank table.

Team Lead: [Name] | Date: [ ]